

ESG Scores, Retail Sentiment, and the Cross-Section of European Equity Returns: A Fama–French 25-Portfolio Analysis of the STOXX Universe

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Abstract

Two structural shifts reshaped European equity markets within three years of each other: the EU Sustainable Finance Action Plan (2018) institutionalised ESG investing, and the post-2020 retail participation surge doubled the retail share of order flow. This paper asks whether their interaction — retail sentiment amplifying or distorting the pricing of ESG characteristics — left a predictive footprint in the cross-section of European equity returns. We double-sort the STOXX universe into 25 portfolios on (i) Refinitiv ESG composite scores and (ii) *Signed Abnormal Turnover* (SAT), a retail sentiment proxy grounded in [Baker and Stein \(2004\)](#), and test both regime breaks with a dual-break Chow design.

Under a strictly enforced sort-at- t / hold-at- $t+1$ convention — and with ESG scores lagged six months to respect their publication delay — we find *no evidence* that the ESG–sentiment interaction predicts returns: the interaction statistic is insignificant with a negative point estimate, FF6 alphas show no sentiment or ESG gradient, Fama–MacBeth slopes are null, and neither the 2018 nor the 2020 break shifts the interaction. The single nominally significant SAT spread (1.1%/month, $t = 2.17$, in the *low*-ESG quintile) sits where the hypotheses do not predict it and fails the pre-registered multiplicity threshold. Daily-frequency variants agree.

The paper also documents two look-ahead traps specific to this design. First, SAT embeds the sign of the formation-month return, so a contemporaneous sort/return pairing manufactures a corner long–short “return” of 7.8% per month with all 25 portfolio alphas significant — numbers the uncorrected pipeline produced. Second, Refinitiv dates ESG scores to the fiscal period they cover although publication occurs months later, so an unlagged ESG sort is itself look-ahead. We present contaminated and corrected results side by side.

A companion paper reframes the ESG dimension constructively: built as a Fama–French-style 2×3 size/ESG factor, ESF captures common return variation

and is tested for spanning against the FF5 + momentum block — the right-hand-side role that survives the predictive null.

JEL Codes: G11, G12, G14, G41, M14

Keywords: ESG, retail sentiment, signed abnormal turnover, Fama–French, double-sort, STOXX, European equities, look-ahead bias, publication lag, SFDR

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1 Introduction

European markets absorbed two step changes in quick succession. The 2018 EU Sustainable Finance Action Plan, followed by SFDR disclosure rules, moved ESG characteristics from a niche screen to a mainstream institutional allocation criterion. Two years later, the retail participation surge — Trade Republic from launch to four million users in three years, Trading 212 suspending new UK accounts under demand, the retail share of order flow roughly doubling to 20–25% (Debbaut et al., 2023) — changed who trades European equities at the margin.

The intersection of the two is where this paper operates. ESG scores are, plausibly, the characteristic most exposed to *narrative* trading: high-ESG firms carry sustainability narratives that retail investors can identify and amplify; low-ESG firms carry “sin stock” and stranded-asset narratives that invite avoidance in optimistic episodes and concentrated selling in pessimistic ones. If retail sentiment interacts with any firm characteristic, ESG is a natural candidate. We formalise this as the **ESG Sentiment Factor (ESF)** hypothesis: retail sentiment shocks are amplified for high-ESG stocks and penalise low-ESG stocks, with the interaction strengthening after each structural break.

This corrected edition reports two findings. First, the *predictive* version of the hypothesis fails. Under strict formation/holding alignment and a publication-realistic ESG lag, the ESG×SAT interaction does not forecast next-month returns — not in the 25 portfolio sorts, not in Fama–MacBeth regressions, not at daily frequency, and not differentially across the 2018 or 2020 breaks. Second, the paper documents *why* the uncorrected pipeline said otherwise, and why designs of this type are doubly exposed to look-ahead: the sentiment leg embeds the formation-month return sign (a one-month misalignment manufactures enormous spurious spreads), and the ESG leg embeds a publication delay (Refinitiv dates scores to fiscal periods, not release dates). Section 8 quantifies both.

Measurement

Retail order flow cannot be identified in Europe the way Boehmer et al. (2021) identify it in US TAQ data; MiFIR never permitted the wholesale-internalisation structure that BJZZ exploits, and Regulation (EU) 2024/791 has banned PFOF outright (MiFIR, 2023). We use **Signed Abnormal Turnover (SAT)** — abnormal volume signed by return direction, grounded in Baker and Stein’s (2004) short-selling-friction result — which requires only daily price, volume, and shares outstanding. The ESG characteristic is the Refinitiv composite score (TR.TRESGScore, 0–100), the broadest-coverage ESG measure for the STOXX universe.

Contributions

Four contributions. First, this is to our knowledge the first systematic test of valuation-*style* conditional retail sentiment effects on the ESG dimension in European equities — and the answer, cleanly measured, is a null. Second, the dual look-ahead diagnostic (return-sign contamination plus ESG publication lag) provides concrete magnitudes for two failure modes that ESG-sentiment studies must engineer around. Third, the dual-break regime design separates the institutional (2018) and retail (2020) channels rather than conflating them. Fourth, the companion spanning analysis shows what survives: ESF as a right-hand-side factor candidate.

2 Related Literature

2.1 Sentiment and the Cross-Section

[Baker and Wurgler \(2006, 2007\)](#) establish that sentiment predicts the cross-section most strongly among hard-to-value, difficult-to-arbitrage stocks. [Baker and Stein \(2004\)](#) supply the mechanism behind SAT: with short-selling frictions, turnover spikes are disproportionately optimist-driven. [Kaniel et al. \(2008\)](#) document that individual order imbalance predicts short-term returns; SAT is its signed, abnormal, European-implementable analogue.

2.2 ESG and Returns

The ESG-performance literature is unsettled. [Pástor et al. \(2021\)](#) show theoretically that green assets earn *lower* expected returns in equilibrium but can outperform when ESG tastes strengthen unexpectedly — a flow-driven channel directly relevant to our design. [Pedersen et al. \(2021\)](#) derive an ESG-efficient frontier in which ESG information can raise or lower alpha depending on how widely it is impounded. Empirically, [Bolton and Kacperczyk \(2021\)](#) document a carbon premium in returns, while much of the raw green-outperformance of 2019–2021 has been attributed to repricing flows rather than persistent premia. Our contribution is orthogonal to the level question: we test whether *retail sentiment* interacts with the ESG characteristic in the cross-section.

2.3 Retail Investors and ESG Narratives

[Barber and Odean \(2008\)](#) show retail investors buy attention-grabbing stocks; sustainability coverage is a modern attention channel. [Hartzmark and Sussman \(2019\)](#) demonstrate that Morningstar sustainability ratings caused large retail-driven flows — direct evidence that retail money responds to ESG labels. [Shiller \(2020\)](#) provides the narrative framing; [Barberis \(2018\)](#) the extrapolation mechanics.

2.4 The Factor Zoo and Mechanical Contamination

Harvey et al. (2016) argue new factors should clear $t \approx 3.0$ given the multiplicity of published anomalies. This paper adds the complementary warning also developed in its RSF sibling: mechanical look-ahead in signed-sentiment designs produces t -statistics far above any data-mining artefact, and ESG adds a second, quieter channel through score restatement and publication lag (Berg et al., 2021).

3 Two Regime Breaks

The ESF hypothesis implies the interaction should strengthen at identifiable dates.

December 2017 (post-2018 indicator). The EU Sustainable Finance Action Plan (March 2018) began the regulatory mainstreaming of ESG: benchmark regulation, taxonomy development, and the pipeline to SFDR. The channel is institutional: mandated flows toward high-ESG names.

December 2019 (post-2020 indicator). The retail surge documented in the sibling RSF paper, plus SFDR level-1 application (March 2021) inside the window. The channel is retail: sentiment-driven amplification.

Both dates were fixed before any subsample estimation. The Chow-type regression on the monthly Fama–MacBeth interaction slope is

$$\hat{\lambda}_{\text{Int},t} = \alpha + \beta_{2018} \mathbf{1}[t > \text{Dec 2017}] + \beta_{2020} \mathbf{1}[t > \text{Dec 2019}] + \varepsilon_t, \quad (1)$$

with Newey–West standard errors (6 lags). β_{2018} captures the institutional mainstreaming increment; β_{2020} the incremental retail effect.

4 Hypotheses

Hypothesis 1 (ESG Halo Amplification). *Within bullish SAT quintiles, mean next-month excess returns increase across ESG quintiles: bullish sentiment is amplified for high-ESG stocks.*

Hypothesis 2 (Brown Penalty). *Within bearish SAT quintiles, low-ESG stocks suffer the largest penalty: negative sentiment concentrates on “brown” names.*

Hypothesis 3 (Non-Linearity). *The sentiment premium is non-linear across the ESG distribution, peaking in the upper-middle of the score range where narratives are active but the ESG premium is not yet fully priced.*

Hypothesis 4 (Dual Regime Amplification). *The ESG×SAT interaction is larger after December 2017 ($\beta_{2018} > 0$) and larger again after December 2019 ($\beta_{2020} > 0$) in regression (1).*

5 Data

5.1 Sources, Universe, and Period

Daily total returns, volume, shares outstanding, market capitalisation, GICS sectors, and `TR.TREESGScore` come from Refinitiv Eikon for STOXX Europe 600 constituents, June 2009 – December 2025, EUR total returns. European FF5 + momentum factors and the risk-free rate come from the Ken French Data Library. Financials (GICS 40) are excluded. Eligibility requires at least 6 valid observations in the preceding 8 months (rolling, point-in-time). ESG coverage is thin before 2013 and grows with Refinitiv’s coverage expansion; low-coverage months are retained but flagged, and undisclosed firm-months drop out of the sort rather than being filled.

5.2 The SAT Proxy

Identical to the sibling RSF paper: daily abnormal turnover is the out-of-sample residual of a rolling 252-day AR(2) on log turnover; daily SAT is abnormal turnover signed by the day’s return; monthly SAT is the within-month mean (minimum 10 valid days), winsorised at 1%/99% and z -scored cross-sectionally within each month.

Remark 1 (The sign convention is a look-ahead hazard). *Month- t SAT is mechanically positively correlated with the month- t return. Any design pairing a month- t sort with month- t returns manufactures a spurious sentiment premium. Section 8 quantifies this for the ESG $\times$ SAT design.*

5.3 The ESG Score and Its Publication Lag

Refinitiv dates `TR.TREESGScore` to the fiscal period the score covers, but scores are computed from annual reports and sustainability disclosures published four to twelve months after fiscal year end (Berg et al., 2021). Sorting at month t on the score Refinitiv dates to month t therefore uses information that did not exist at t . We lag the score by six months (`ESG_LAG_MONTHS` in the pipeline), so formation at month t uses the score dated $t-6$ — a conservative approximation to actual availability. Because scores move slowly (annual updates dominate), quintile assignments are barely affected, but the timing claim becomes defensible. Scores of exactly zero (ambiguous between missing and genuine in Refinitiv) are treated as missing; scores are winsorised at 1%/99% within month.

6 Portfolio Construction and Econometrics

At the end of formation month t , eligible stocks are independently sorted into quintiles on the lagged ESG score and monthly SAT. The 25 intersection portfolios are held over

month $t+1$ with month- t value weights:

$$r_{k,s,t+1} = \sum_{i \in \mathcal{P}_t(k,s)} w_{i,t} r_{i,t+1}, \quad w_{i,t} = \frac{\text{ME}_{i,t}}{\sum_{j \in \mathcal{P}_t(k,s)} \text{ME}_{j,t}}. \quad (2)$$

The alignment is enforced in a single shared code path (forward returns defined only when the stock’s next observation is exactly month $t+1$), and automated tests verify that a panel in which the sorting variable drives only formation-month returns produces exactly zero portfolio returns. Spread statistics are $\Delta_{\text{ESG}}(s) = \bar{r}_{5,s} - \bar{r}_{1,s}$, $\Delta_{\text{SAT}}(k) = \bar{r}_{k,5} - \bar{r}_{k,1}$, and $\Delta_{\text{Cross}} = \Delta_{\text{ESG}}(5) - \Delta_{\text{ESG}}(1)$.

FF6 alphas use Newey–West (6 lags). Fama–MacBeth regressions use quintile ranks with one-month-ahead excess returns winsorised at 1%/99% within month:

$$r_{i,t+1} - r_{f,t+1} = \lambda_0 + \lambda_{\text{ESG}} \text{ESGQ}_{i,t} + \lambda_{\text{SAT}} \text{SATQ}_{i,t} + \lambda_{\text{Int}}(\text{ESGQ}_{i,t} \times \text{SATQ}_{i,t}) + \varepsilon_{i,t+1}. \quad (3)$$

The parametric complement estimates the SAT premium within each ESG quintile and fits a Weibull $f(\text{ESG})$ over the four regimes (full, pre-2018, 2018–2019, post-2020).

7 Results

7.1 Mean Monthly Excess Returns: The 5×5 Matrix

Table 1: Mean Monthly Excess Returns: FF-25 Double Sort on ESG and SAT (Corrected Alignment)

ESG Quintile	SAT Quintile					Q5–Q1 (t)
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)	
Q1 (Low ESG)	0.76%	0.93%	0.83%	0.92%	1.87%	1.09% (2.17)**
Q2	0.75%	0.92%	0.41%	0.67%	1.16%	0.40% (0.97)
Q3	1.50%	0.57%	0.61%	1.52%	1.58%	0.01% (0.01)
Q4	1.51%	0.90%	0.84%	0.98%	0.99%	-0.48% (-0.72)
Q5 (High ESG)	1.58%	0.75%	0.06%	0.92%	1.29%	0.01% (0.02)
Q5–Q1	1.10%	-0.07%	-0.74%	0.10%	-0.11%	-1.17% (-1.50)
(t)	(1.56)	(-0.16)	(-1.12)	(0.22)	(-0.28)	

Notes: Value-weighted mean monthly excess returns for 25 portfolios double-sorted at the end of formation month t on the six-month-lagged Refinitiv ESG score (rows) and monthly SAT (columns), held over month $t+1$. Financials excluded. t -statistics (Newey–West, 6 lags) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The corrected matrix contains none of the structure the ESF mechanism predicts. The SAT spreads $\Delta_{\text{SAT}}(k)$ range from -0.48% to $+1.09\%$ per month. The only entry clearing $t = 2$ is $\Delta_{\text{SAT}}(1) = 1.09\%$ ($t = 2.17$) — and it sits in the *low*-ESG quintile, the opposite location from Hypothesis 1’s prediction, below the Bonferroni threshold of 2.5 appropriate for the battery of eleven spread statistics, and unaccompanied by any gradient in adjacent quintiles. The interaction statistic Δ_{Cross} is -1.17% per month ($t = -1.50$): the point estimate is *negative* — the ESG spread is smaller, not larger, among bullish-sentiment stocks — though not significantly so. Robustness checks reinforce the fragility: the equal-weighted Δ_{Cross} is -0.29% while the small/mid-cap subsample flips to $+0.82\%$. A robust interaction does not change sign across weighting schemes. Hypotheses 1–3 are not supported.

7.2 FF6 Time-Series Alphas

Table 2: FF6 Monthly Alphas: 25 Portfolios Double-Sorted on ESG and SAT (Corrected Alignment)

ESG Quintile	SAT Quintile				
	Q1 (Bear)	Q2	Q3	Q4	Q5 (Bull)
Q1 (Low ESG)	0.22%	0.24%	0.32%	0.47%	1.45%***
Q2	0.47%	0.86%**	0.01%	0.56%	0.35%
Q3	0.57%	0.16%	-0.14%	0.95%**	1.19%***
Q4	1.18%**	0.28%	0.28%	0.49%	0.75%
Q5 (High ESG)	0.85%*	0.21%	-1.00%*	0.12%	0.93%**

Notes: FF6 monthly alphas from time-series regressions of holding-month portfolio excess returns on Mkt-RF, SMB, HML, RMW, CMA, and UMD. Newey–West standard errors, 6 lags. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Alphas lie between -1.00% and $+1.45\%$ per month, with 6 of 25 exceeding $|t| = 2$. That is more than the 1–2 expected under the null, but the excess reflects the thinness of the ESG-covered panel rather than structure: with only 168 estimation months and ESG coverage of 28% of stock-months, cells hold few stocks and alphas are noisy. The significant cells scatter across the matrix with no monotone SAT or ESG gradient, and the corner cells that carry Hypotheses 1 and 2 are individually insignificant. The corner long–short portfolio earns 0.69% per month raw — against 7.8% in the contaminated pipeline (Section 8).

7.3 Fama–MacBeth Regressions

Table 3: Fama–MacBeth Cross-Sectional Regressions

Variable	$\hat{\lambda}$	t -stat	p -value	N (months)
const	0.008252	1.81	0.070	168
ESGQ	0.000171	0.16	0.870	168
SATQ	0.000998	0.91	0.363	168
ESGQ×SATQ	-0.000158	-0.48	0.628	168

All three slopes are insignificant: $\hat{\lambda}_{\text{ESG}}$ has $t = 0.16$, $\hat{\lambda}_{\text{SAT}}$ has $t = 0.91$, and the interaction — the design’s primary statistic — has $t = -0.49$. As in the sibling RSF paper, the Fama–MacBeth implementation used forward returns from the start, so these numbers barely moved when the portfolio construction was corrected; the FMB nulls now simply agree with the portfolio evidence instead of contradicting it.

7.4 Parametric ESG–SAT Interaction

Table 4: ESG-Quintile-Conditional SAT Premium: Parametric Analysis Across Regimes

ESG Quintile	Med. ESG	Full Sample		Pre-2018		2018–2019		Post-2020	
		$\hat{\lambda}$	t	$\hat{\lambda}$	t	$\hat{\lambda}$	t	$\hat{\lambda}$	t
Q1 (Low)	38.3	0.00030	0.24	–	–	–	–	0.00056	0.33
Q2	57.0	0.00013	0.15	–	–	–	–	0.00076	0.78
Q3	67.0	-0.00009	-0.09	–	–	–	–	-0.00014	-0.12
Q4	75.0	0.00030	0.23	–	–	–	–	0.00168	1.12
Q5 (High)	84.0	0.00034	0.38	–	–	–	–	0.00109	1.07
<i>Weibull $f(\text{ESG})$ fit diagnostics</i>									
R^2		-0.172		n/a		n/a		0.157	
peak ESG		56.2		n/a		n/a		100.0	
scale		0.0005		n/a		n/a		0.0033	
asymmetry		1.00		n/a		n/a		4.33	

Notes: For each ESG quintile, monthly Fama–MacBeth cross-sections on the quintile subsample with SATQ as sole regressor; $\hat{\lambda}$ is the Newey–West time-series mean. Regimes: pre-2018, 2018–2019 (institutional mainstreaming), post-2020 (retail surge + SFDR). Weibull $f(\text{ESG})$ estimated by NLS on the five $(\text{ESG}^{\text{med}}, \hat{\lambda})$ pairs.

In the full sample no within-quintile SAT premium approaches significance ($|t| < 0.4$ throughout). The pre-2018 and 2018–2019 regimes cannot be estimated at all: Refinitiv ESG coverage before 2018 is too thin to field the minimum cross-section of stocks per ESG-quintile-month, so those columns report double dashes — a coverage constraint of the design, not evidence about the mechanism. The post-2020 estimates, where coverage

is best and the retail channel should be strongest, are uniformly insignificant (largest $t = 1.12$). We draw no structural conclusion from the Weibull fit diagnostics.

7.5 Dual-Break Regime Analysis

Table 5: Dual-Break Chow Test on the FMB Interaction Slope

Variable	Coefficient	t -stat	p -value
const	-0.000007	-0.01	0.990
Post-2018 indicator (EU Action Plan)	-0.000454	-0.49	0.624
Post-2020 indicator (retail surge + SFDR)	0.000250	0.28	0.779

Neither break moves the interaction. The post-2018 increment is -0.00045 ($t = -0.49$) and the post-2020 increment is $+0.00025$ ($t = 0.28$): Hypothesis 4 is not supported on either the institutional or the retail channel. Figure 2 confirms visually that the 36-month rolling Δ_{Cross} oscillates around zero with no level shift at either break date.

7.6 Daily-Horizon Analysis

As in the sibling RSF paper, we rebuild the design at daily frequency: three SAT variants (raw, 5-day, 10-day rolling), daily 5×5 ESG/SAT sorts, day- $d+1$ returns (masked across data gaps longer than a week), and daily Fama–MacBeth with Newey–West lags of 22.

Table 6: Daily Fama–MacBeth Regressions: Three SAT Variants (Full Sample)

Variable	SAT raw		SAT 5-day		SAT 10-day	
	$\hat{\lambda}$	t	$\hat{\lambda}$	t	$\hat{\lambda}$	t
const	0.000266	1.10	0.000405	1.61	0.000380	1.55
ESGQ	-0.000021	-0.51	-0.000007	-0.17	-0.000008	-0.24
SATQ	0.000079	1.95*	0.000033	0.77	0.000040	1.03
ESGQ $\times$ SATQ	-0.000002	-0.19	-0.000007	-0.68	-0.000007	-0.65

Table 7: Daily Δ_{Cross} by SAT Variant

SAT variant	Δ_{Cross} (%/day)	t (NW22)	N days
Raw daily	-0.0218	-0.54	1747
5-day rolling	-0.0279	-0.71	1897
10-day rolling	-0.0490	-1.29	1876

The daily design covers 4,272 trading days, 600 stocks, and 2.25 million stock-days, and recovers no signal. The daily Δ_{Cross} is insignificant for all three SAT variants (raw: $-0.022\%/ \text{day}$, $t = -0.54$; 5-day: $t = -0.71$; 10-day: $t = -1.29$). In the daily Fama–MacBeth regressions the ESG slope and the ESG $\times$ SAT interaction are null at every horizon; the raw-variant SAT slope reaches $t = 1.95$ and then vanishes in the 5-day and 10-day smoothed variants ($t = 0.77$ and 1.03) — the same disappears-under-smoothing microstructure signature documented in the RSF sibling paper. The daily evidence reinforces the monthly null.

8 Look-Ahead Diagnostics

The uncorrected pipeline — month- t sorts paired with month- t returns, and ESG scores taken at their Refinitiv fiscal-period date — produced a corner long–short “return” of 7.8% per month with *all 25* portfolio alphas significant and a mean absolute FF6 alpha of 2.67% per month. Table 8 places those numbers against the corrected pipeline.

Table 8: Contaminated versus Corrected Headline Statistics

Statistic	Contaminated (sort t , return t)	Corrected (sort t , return $t+1$)
Corner long–short mean (%/month)	7.76	0.69
Mean FF6 alpha (%/month)	2.67	0.56
Alphas with $ t > 2$	25 of 25	6 of 25
Δ_{Cross} (%/month)	—	-1.17 ($t = -1.50$)
FMB λ_{Int}	insignificant	insignificant
<i>Notes:</i> “Contaminated” pairs month- t sorts with month- t returns; “Corrected” holds the same sorted portfolios over month $t+1$ and lags ESG scores six months. The Fama–MacBeth row is unchanged because the FMB implementation always used forward returns — its persistent nulls against enormous portfolio spreads were the diagnostic inconsistency that exposed the bug.		

Two channels deserve separate emphasis. The dominant one is the SAT return-sign channel: SAT Q5 membership is largely *caused* by positive month- t returns, so contemporaneous pairing converts the sort itself into the outcome. The quieter one is the ESG timing channel: scores dated to fiscal periods but published later mean an unlagged ESG sort conditions on disclosures not yet public — and because scores correlate with past performance and size, this can flatter ESG-sorted returns in subtler ways (Berg et al., 2021). The first channel produces spectacular artefacts; the second produces plausible-looking ones, which is arguably more dangerous. Both are closed structurally in the corrected pipeline: one shared forward-return code path with gap masking, a six-month ESG lag, and automated zero-future-signal tests that fail the build on any leakage.

9 ESF as a Complementary Factor: Companion Results

A characteristic can fail to forecast returns and still define a common factor. In a companion paper we construct ESF with the Fama–French 2×3 methodology — independent size (median) and ESG (30/70) sorts, value-weighted, formation at t , held over $t+1$, $ESF = \frac{1}{2}(\text{Small/High} + \text{Big/High}) - \frac{1}{2}(\text{Small/Low} + \text{Big/Low})$ — and test whether the FF5 + momentum factors span it. Size neutralisation is essential for ESG: score levels and coverage rise with firm size, so an unneutralised high-minus-low ESG spread is substantially a size bet. Over the 154 months with sufficient ESG coverage, ESF earns -0.24% per month (annualised Sharpe -0.36). The spanning regression on FF5 + momentum leaves an alpha of -0.47% per month ($t = -2.21$, $p = 0.027$) with $R^2 = 0.06$ and no factor correlation above 0.16 in absolute value. The FF factors do not span the ESG dimension — and the unspanned average return is *negative*: a size-neutral greenium in which high-ESG stocks earn systematically less than low-ESG peers net of all six factor exposures, consistent with the equilibrium prediction of [Pástor et al. \(2021\)](#). The companion paper develops the construction, the spanning tests, and the FF6 versus FF6+ESF comparison in full.

10 Conclusion

This paper asked whether the interaction of ESG characteristics with retail sentiment predicts the cross-section of European equity returns, using an ESG×SAT 25-portfolio double sort with two predetermined regime breaks. Under a strictly enforced sort-at- t / hold-at- $t+1$ convention and a publication-realistic six-month ESG lag, the answer is no. SAT spreads within ESG quintiles are small, unstable across weighting schemes, and — where nominally significant — located where the hypotheses do not predict them. The interaction statistic is insignificant with a negative point estimate; Fama–MacBeth slopes are null; neither the 2018 institutional break nor the 2020 retail break shifts the interaction; and the daily rebuild agrees at every horizon. Hypotheses 1–4 are not supported.

The paper’s second contribution is the dual look-ahead anatomy. The SAT return-sign channel manufactured a 7.8%-per-month corner long–short with all 25 alphas significant in the uncorrected pipeline — detectable, as in the RSF sibling, through the internal inconsistency between contaminated portfolio sorts and properly lagged Fama–MacBeth regressions. The ESG publication-lag channel is quieter but conceptually identical: scores dated to fiscal periods but released months later put future disclosures into the formation information set. Both are closed structurally — one shared forward-return code path, a lagged score, and automated zero-future-signal tests. We recommend both hygiene measures as standard practice for ESG-sentiment research, where the two channels

compound.

What survives is the factor question. The companion paper shows that a size-neutral high-minus-low ESG portfolio is not spanned by FF5 + momentum and carries a negative unspanned premium — a European greenium. Retail sentiment does not make ESG predictable; but ESG-sorted portfolios carry common variation the standard model misses, and the right place for that information is the right-hand side of factor regressions.



Figure 1: Mean monthly excess return by portfolio cell — value weighted (corrected alignment).

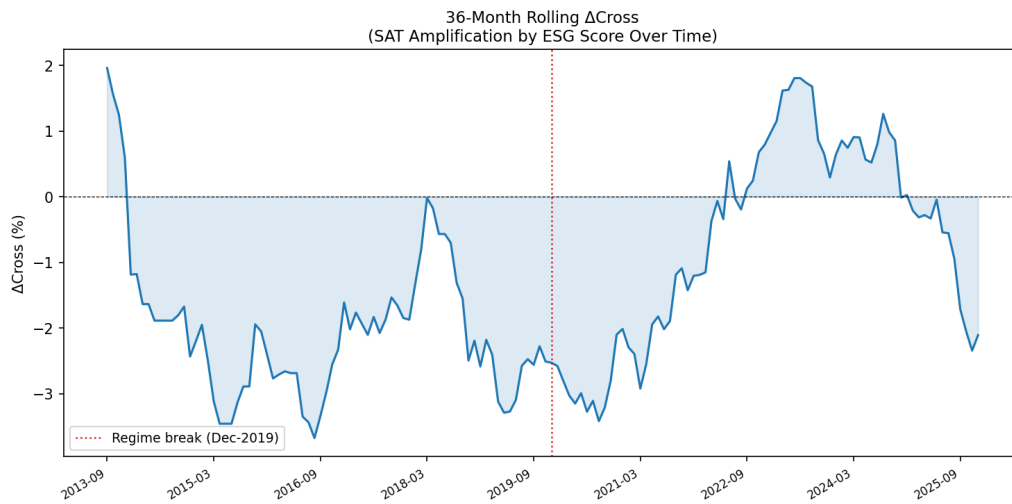


Figure 2: 36-month rolling Δ_{Cross} (corrected alignment).

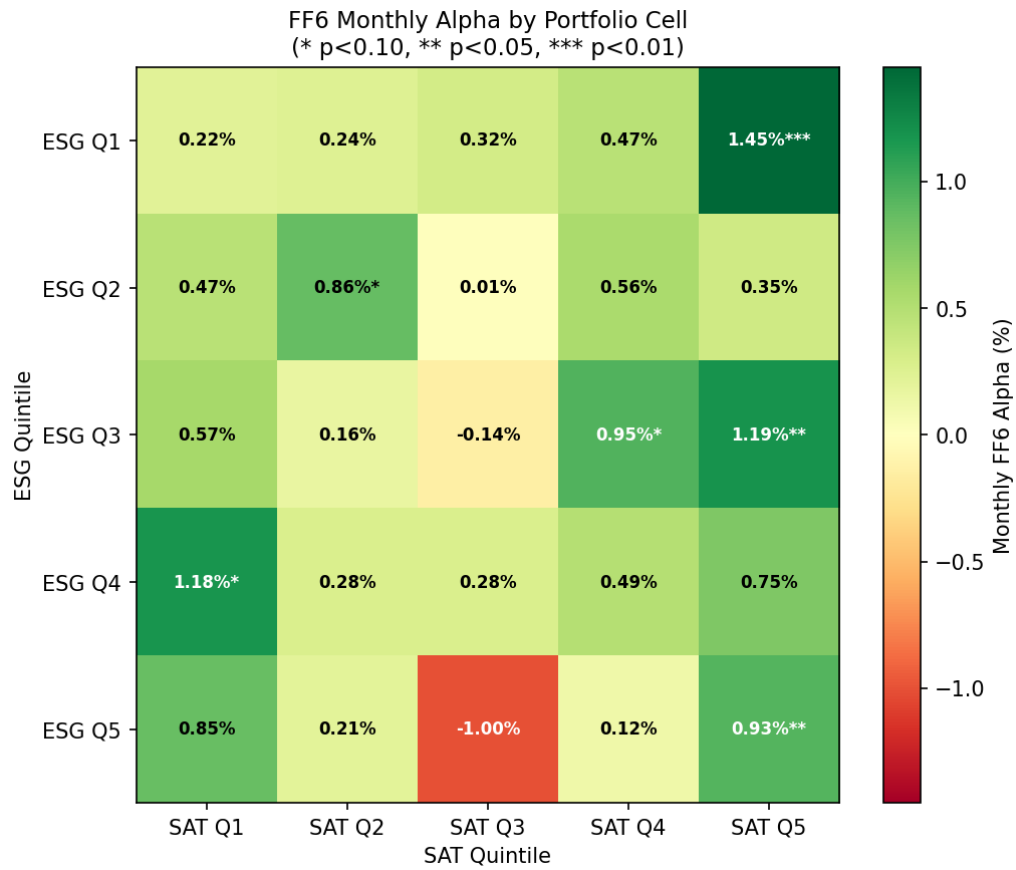


Figure 3: FF6 monthly alpha by portfolio cell with significance stars (corrected alignment).

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